

Report Biomimicry Challenge

ZHAW Institute for Sustainable Development | Blockweek Bionics 2026

Challenge 1 — Urban Heat Islands

Cooling the City



Concept V2 on the reference block: timber ribs and collector panels on the façade (render: project team)

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1 Introduction

1.1 Background & Relevance

The Urban Heat Island (UHI) effect represents a growing environmental and public health challenge, characterized by significantly elevated temperatures in urban environments relative to surrounding rural areas [1]. This effect is worsened by climate change and a high concentration of heat absorbing infrastructure, like cement and asphalt. Conventional mitigation strategies rely heavily on air conditioning, which creates a counterproductive cycle. It increases external temperatures through heat rejection and elevates the energy consumption, often causing higher greenhouse gas emissions.

To address this problem, the report investigates nature inspired passive cooling mechanisms, also known as biomimicry.

1.2 Design Challenge & Project Contribution

The design challenge lies in developing an efficient, scalable cooling system capable of mitigating the urban heat island effect without relying on high energy consumption.

The goal is to analyze the different heat dissipation methods found in nature, specifically in animals, to find a solution which conforms with the given challenges.

2 Challenge & Scoping

2.1 Problem Definition

Conventional cooling methods such as air conditioning worsen the Urban Heat Island effect by displacing the warm air from inside the building to the surrounding area. This local heat exchange causes the city to warm up more, increasing the need for air conditioning, which ultimately results in a self-reinforcing (positive) feedback loop.

2.2 Context and Constraints

All context values, benchmarks and engineering checks in this report refer to the City of Zurich. Fixing one reference city is what makes the constraints and the KPI targets checkable. Zürich was chosen for four reasons:

1. The city has adopted a heat-mitigation strategy, the Fachplanung Hitzeminderung [2], with an implementation agenda, a guidance leaflet for building owners and thematic maps. The project therefore has a documented baseline and published target values instead of invented ones.
2. The city publishes an open climate analysis including geodata, so the reference site can be selected from measured data instead of intuition.
3. MeteoSwiss operates long term measurement stations in and around the city.
4. Its building style and infrastructure are representative of the Swiss Mittelland, so results transfer to comparable Swiss cities.

2.2.1 Reference Building

For comparison and reference purposes a building inside of Zurich was chosen. The chosen building is the Modellierungsgebiet 03 – Geschlossene Randbebauung (closed perimeter block). It consists of two outward-closed blocks of five stories including a roof structure and an enclosed private courtyard. It is located in a dense district close to the city center. The city’s assessment is that the reduced air exchange and lack of cold air supply makes this building overheat comparatively quickly.

2.2.2 Constraints

Table 2.1 separates hard constraints, where a violation disqualifies a concept, from soft constraints, which weight the evaluation but do not necessarily disqualify a concept.

Table 2.1 – Context conditions and constraints.

Type	Class	Constraint
Energy	hard	Limit auxiliary electrical energy for the cooling function.
Water	hard	No unnecessary water waste.
Townscape	hard	The result must be acceptable in the Zurich townscape. Buildings may be listed; heritage protection, building law and neighbor rights can restrict visible changes [2], [3, p. 128].
Glare	hard	No specular glare towards streets, neighboring flats or traffic, which can impede day to day life.
Climate	hard	Must survive the Swiss annual cycle: frost to about -10°C , freeze–thaw cycles, driving rain, UV-light, and periods of several dry weeks in summer.
Winter	hard	Must not increase heating demand. A high solar reflectance also reduces useful winter solar gains; this trade-off must be quantified, not ignored.
Existing building	soft	Preferable, but not necessary. Concepts can also be developed for only new buildings.
Installation	soft	Must be installable by ordinary trades on an occupied building, ideally without full scaffolding.
Maintenance	soft	Must remain functional with little maintenance.
Solar energy	soft	Should not unnecessarily compromise roof- or façade-integrated photovoltaics.
Structural	soft	Limited additional dead load and build-up thickness on an existing envelope.
Embodied emissions	soft	The greenhouse gas embodied in the added infrastructure must be repaid by the avoided cooling demand within the service life, measured in $\text{kg CO}_2\text{-eq per m}^2$.
Economic	soft	Cost per square meter should be in the range of a normal refurbishment.

2.3 System View and Project Boundary

2.3.1 Super-system

The chosen super system is the city and its existing infrastructure.

2.3.2 System of interest

The core interest of this project focuses on heat-regulation of the outer layer of a building including the façade and the roof.

2.3.3 Parallel systems

There are several parallel systems which might need to be involved or considered when finding solutions for the system of interest:

- Stormwater and sewer network
- Street and pavement network
- Power grid and building energy system
- Residents
- Cars, bicycles, public transport and other forms of transportation

2.3.4 Subsystems

There are several subsystems involved in the system of interest. These subsystems are to be considered when developing a concept.

- Outer surface of a façade
- Insulation layer
- Water paths
- Attachment to the substrate

Table 2.2 states which aspects are scoped in or out to further narrow down the project boundaries.

Table 2.2 – Project boundary: what is deliberately in and out of the project.

IN the project	OUT of the project
Prevent heat buildup and remove heat from the building façade.	Redesign of the building itself. Reuse of the stored energy. No internal cooling.

2.4 Design Question

2.4.1 Candidate questions considered

To define the project several design questions were formulated and evaluated. Table 2.3 shows the rejected candidates.

Table 2.3 – Design questions considered, and why they were rejected.

#	Candidate	Assessment
1	How do we prevent heat build-up in urban areas?	Too broad. No place, no time, no intervention point. The equivalent of “How might we end world hunger?”
2	How could we use the energy or heat build-up positively while lowering temperature?	Right idea, but unanchored: no location, and “positively” does not say for what.
3	How might we get the heat that Zurich’s existing buildings store during the day back out of the city before the next morning, without using energy?	Well-formed and highly testable. Needs only small adjustment. But also a bit too narrow and does not leave a lot of variation.
4	How can we prevent heat waves causing heat islands?	Factually wrong. Heat waves do not cause heat islands. The urban heat island is a permanent consequence of urban infrastructure and exists on ordinary summer days. A heat wave makes it dangerous, but it does not create it.

2.4.2 Selected design question

How might we reduce the urban heat island effect, in cities like Zurich?

This question is formulated neither too narrowly nor too widely. It allows some freedom in how to reduce the urban heat island effect, but it also refrains from a too open interpretation by narrowing the scope of the project to cities similar to Zurich.

2.5 Functions and Functional Diagram

2.5.1 Functions

Table 2.4 displays the chosen functions and their respective taxonomy link.

Table 2.4 – Functions list.

ID	Function (verb)	Taxonomy link – Explanation
F1	Reflect short-wave radiation	Prevent solar radiation – Avoid absorbing solar energy (visible & near infrared light).
F2	Shade a surface	Prevent excess temperature – Reduce direct light exposure on structural surfaces.
F3	Dissipate / transfer heat	Modify thermal conditions – Transfer thermal energy via transmission or thermal radiation.
F4	Evaporate liquid	Regulate physical processes – Heat extraction using moisture or water.
F5	Resist degradation	Prevent structural breakdown – Maintain long-term durability against UV, rain, and frost.

2.5.2 Functional Diagram

Using the functions listed in Table 2.4 a couple of flows can be constructed (Figure 2.1).

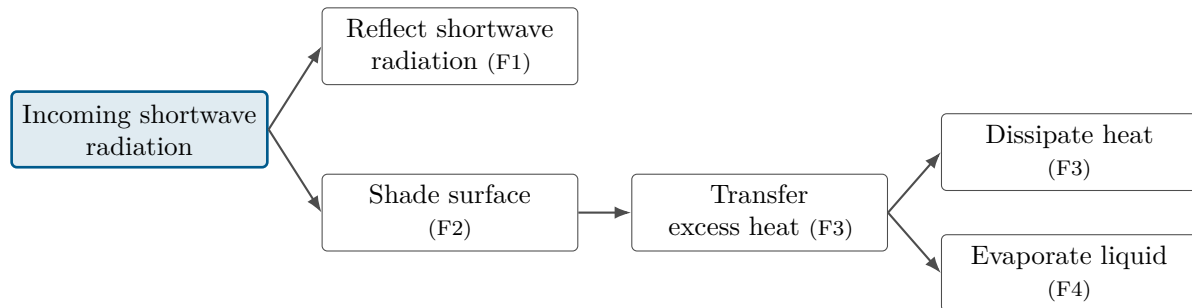


Figure 2.1 – Functional diagram: the two flows built from the functions of Table 2.4. Flow 1 reflects the incoming radiation; flow 2 shades the surface and transfers the excess heat, which is then dissipated or removed by evaporation.

2.6 Biologized Questions

To ease the search for solutions developed by nature a few biologized questions were formulated:

1. How does nature reflect incoming shortwave radiation?
2. How does nature shade a surface even in the desert?
3. How does nature dissipate excess heat?
4. How does nature transfer excess heat?
5. How does nature dissipate excess heat using moisture?
6. How does nature keep a functional surface clean without cleaning it?

2.7 Performance Criteria and KPIs

2.7.1 KPIs

Table 2.5 shows the measurable KPIs used as benchmarks for the project.

Table 2.5 – KPIs.

#	Function	KPI	Unit
1	F1	Reflection index using the albedo index [4]	0 – 1
2	F2	Time a surface is in the shade [5]	m ² /h spent in the shade
3	F3	Heat dissipated	kW
4	F3	Heat transferred	kW
5	F4	Water evaporated	L
6	F5	Efficiency factor after a set amount of time	% of initial

2.7.2 Life's Principles

The Life's Principles which should reflect the end solution include:

- Be resource efficient: minimize the use of external resources
- Adapt to changing conditions: adapt to day/night time as well as season cycles
- Use life-friendly chemistry: refrain from the use of toxic chemicals or materials

3 Bio-Inspired Design Process

3.1 Discovering & Abstracting

3.1.1 Research path

To find suitable solutions both the AskNature website as well as the AskNature chat were used. The following keywords were used to find solutions produced in nature:

Cooling, thermal transfer, shading, heat reduction, reflect, evaporation

Additionally the biologized questions were used to search for solutions using search engines such as Google.

The models that were selected had to fulfill at least one function mentioned in Section 2. Additionally they had to fit into at least one of the defined Life's Principles. Therefore some models were abandoned and not investigated further.

3.1.2 Natural Models

African Elephant: Ears

Biological mechanism As seen in Figure 3.1, African elephants have large and highly vascularized ears which they use as heat exchange surfaces. Warm blood is transported from the body core into the ear, where heat is transferred to the surrounding air. By flapping their ears additional airflow is created which further increases the heat transfer [6]. This same approach is also used by the jackrabbit [7].

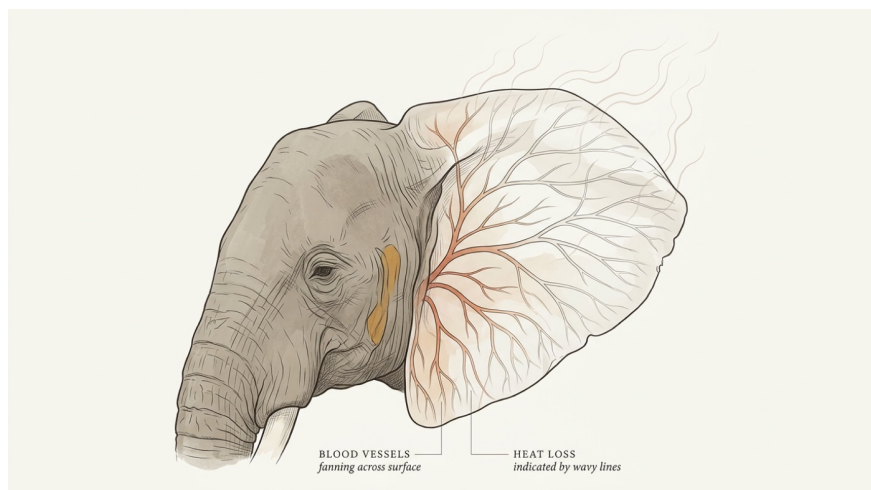


Figure 3.1 – African elephant ears (schematic illustration).

Functional abstraction. Transfer thermal energy from a heat generating region to a large exchange surface and increase airflow across this surface.

Design principle for urban cooling. Use dedicated heat rejection surfaces in locations with favorable airflow to transfer and dissipate heat from urban structures.

Life's Principle: Be resource efficient / Adapt to changing conditions

Elephant Skin: Sparse Hairs

Biological mechanism African elephants have additionally to their ears a second cooling mechanism. They have very sparsely spread hair all over their body. Due to the low density of the body hair it does not act as an insulation layer but rather as additional surface area to transfer heat. As seen in Figure 3.2, it also serves to break the stationary air surrounding the skin. This allows for a better heat transfer [8].

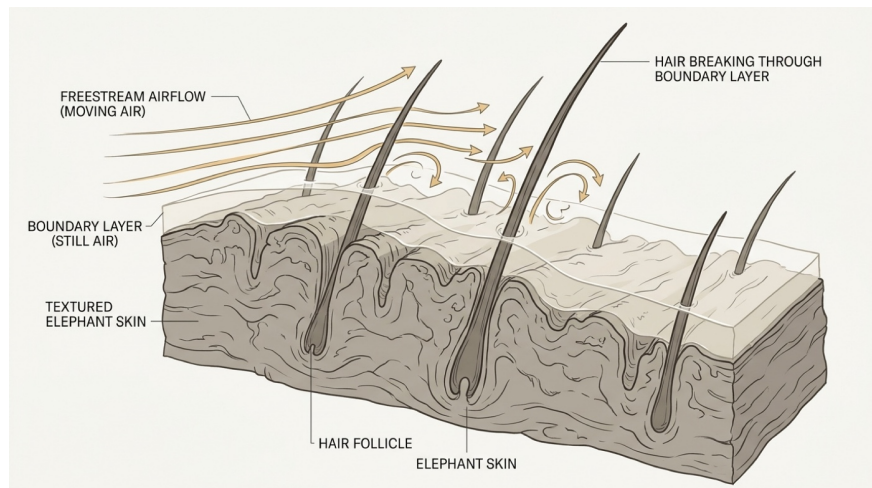


Figure 3.2 – African elephant hair (schematic illustration).

Functional abstraction. Increase heat-transfer area by distributing sparse bulges across a warm surface.

Design principle for urban cooling. Use small surface structures on façades or other heated urban surfaces to enhance passive heat transfer without substantially increasing material use.

Life's Principle: Be resource efficient

Cactus Ribs

Biological mechanism Many cacti have ribbed surfaces consisting of repeated peaks and troughs. The peaks shade parts of the troughs and therefore reduce direct solar exposure. Air within the shaded troughs remains cooler and absorbs heat from the cactus which rises as it warms. The ribbed geometry also increases surface area and disrupts airflow around the plant, similar to the elephant hair [9], [10].

Functional abstraction. Use folded surface geometry to combine self-shading, increased heat-transfer area and controlled interaction with airflow.

Design principle for urban cooling. Integrate ribbed or folded geometries into urban surfaces to reduce direct solar exposure while simultaneously promoting passive heat dissipation.

Life's Principle: Be resource efficient

Saharan Silver Ant: Reflective Hairs

Biological mechanism The Saharan silver ant is covered with densely packed hairs that have a triangular cross-section and grooved surfaces. These hairs reduce heat gain in two ways: they strongly reflect visible and near-infrared solar radiation through total internal reflection and they increase emissivity in the mid-infrared range, allowing the ant to radiate excess body heat more efficiently. AskNature reports that the hairs alone keep the ant about 2 °C cooler, while the combined mechanisms provide an additional cooling effect of approximately 5–10 °C [11].

Functional abstraction. Use micro structured surfaces to reflect incoming solar radiation while simultaneously enhancing thermal radiation away from the surface.

Design principle for urban cooling. Develop façade or roof surfaces with micro structured geometries that minimize absorption in the visible and near-infrared solar spectrum while maximizing heat emission in the mid-infrared. This could reduce solar heat gain and improve passive radiative cooling without requiring active energy input.

Life's Principle: Adapt to changing conditions / Use life-friendly chemistry

Desert Snail Shell

Biological mechanism The desert snail *Sphincterochila boissieri* limits heat gain through a highly reflective shell and internal insulating air spaces, as seen in Figure 3.3. AskNature reports approximately 90 % reflectance in the visible spectrum and about 95 % in the near-infrared range. The snail also withdraws into the upper parts of its shell, while air-filled spaces and limited contact with the hot substrate reduce conductive heat transfer [12].

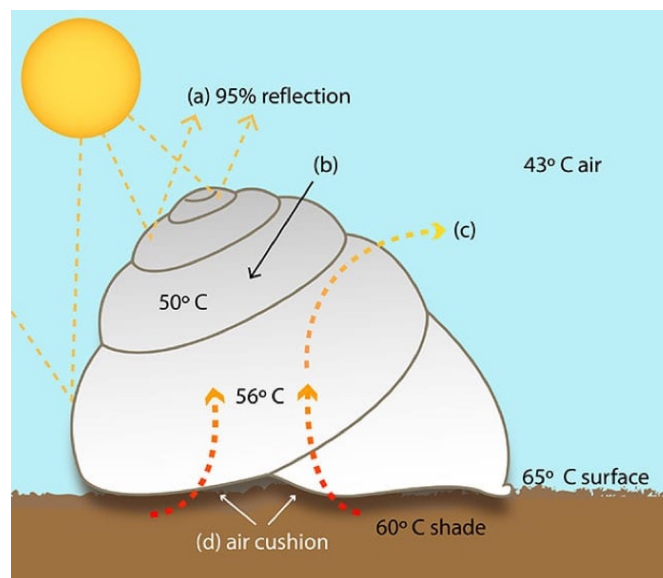


Figure 3.3 – Desert snail [12].

Functional abstraction. Reduce incoming radiant energy and interrupt conductive heat transfer using reflective surfaces and stationary air layers.

Design principle for urban cooling. Combine highly reflective external surfaces with insulating or ventilated air cavities to limit solar heat storage in roofs and façades.

Life's Principle: Be resource efficient / Adapt to changing conditions

Morpho Butterfly Wing

Biological mechanism Morpho butterflies generate their characteristic color using transparent chitin-and-air structures at micro- and nanoscale. As seen in Figure 3.4, ridges and cross-ribs interact with incoming light through diffraction and interference, causing selected wavelengths to be strongly reflected. The color therefore results primarily from physical structure rather than pigmentation [13].

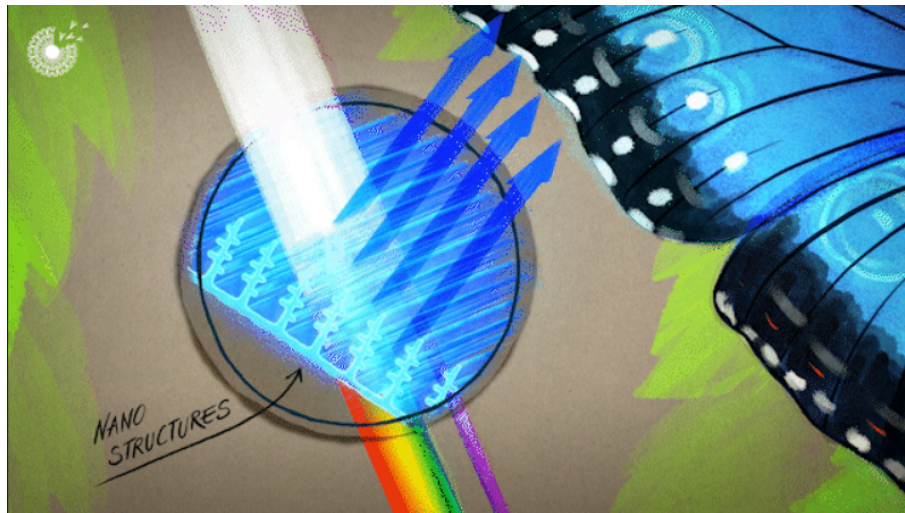


Figure 3.4 – Reflection with nano structures [14].

Functional abstraction. Control the spectral interaction between a surface and incoming radiation through micro- and nanoscale geometry.

Design principle for urban cooling. Engineer structured surfaces that selectively manipulate incident solar radiation while maintaining desired optical properties. It needs to be mentioned that the Morpho demonstrates structural color rather than near-infrared heat rejection and that a cooling application requires separate verification.

Life's Principle: Be resource efficient

Camel Fur and Sweat

Biological mechanism Dromedary camels combine insulation with evaporative cooling. Figure 3.5 displays how exactly this mechanism works. The sweat glands release water which removes the body heat through evaporation, while the thick fur blocks the hot air from reaching the skin. Furthermore the camel reflects light with its pale color [15].

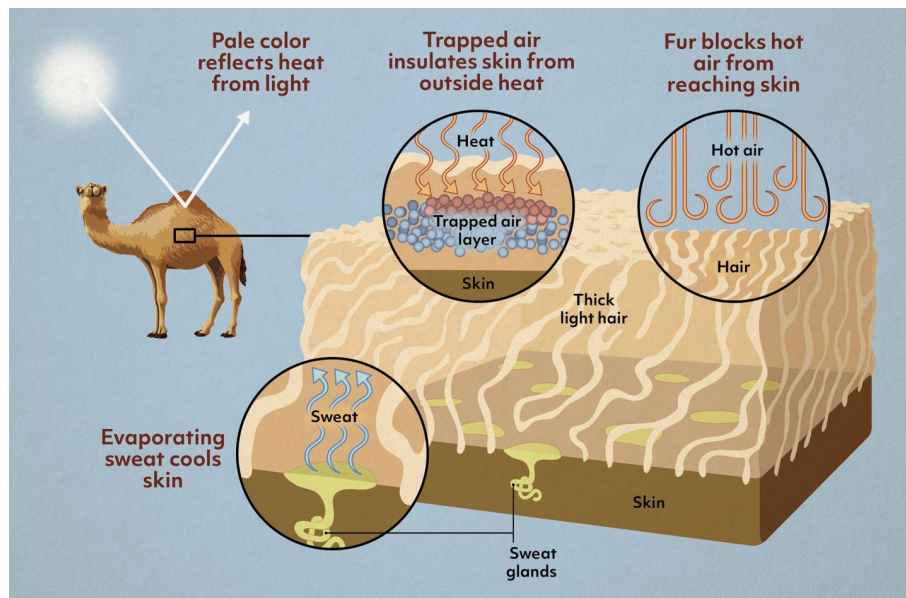


Figure 3.5 – Insulation with evaporative cooling [15].

Functional abstraction. Block heat using insulation in combination with evaporation as a means of heat dissipation.

Design principle for urban cooling. Combine shading or thermal insulation with controlled evaporative cooling so that less water is required to achieve the desired cooling effect.

Life's Principle: Be resource efficient / Adapt to changing conditions / Use life-friendly chemistry

Pine Cone Scales

Biological mechanism Pine cone scales respond to humidity despite containing no muscles or active control system. Their layered structure contains materials with different swelling behavior. When moisture is absorbed, one layer expands more than the other, causing the scale to bend and close. As the scale dries, it returns towards its open state. The movement is reversible [16].

Functional abstraction. Convert an environmental change directly into mechanical movement through differential material expansion.

Design principle for urban cooling. Use environmentally responsive material layers to create self-actuating shading or ventilation elements that operate without motors or electronic control.

Life's Principle: Adapt to changing conditions / Use life-friendly chemistry

Honeybee Fanning Thresholds

Biological mechanism Western honeybee colonies regulate nest temperature through the combined behavior of many workers. Individual bees have different temperature-response thresholds. As nest temperature increases, progressively more bees begin ventilating the hive. This distributed response prevents excessive system-wide reactions to small temperature fluctuations and stabilizes the nest with no central control [17].

Functional abstraction. Distribute different activation thresholds across multiple independent elements to create a gradual system-level response.

Design principle for urban cooling. Use multiple passive cooling elements with different trigger conditions so that cooling capacity increases progressively with thermal load instead of operating through a single on/off threshold.

Life's Principle: Be resource efficient / Adapt to changing conditions

3.2 Creating: Engineering Concept Development

3.2.1 Structured Ideation: Crazy 8s

Table 3.1 translates the biological mechanisms in Section 3.1 into eight technical building blocks.

Table 3.1 – Crazy 8s building blocks, before concept combinations.

Idea	Function	Biological starting point	Technical building block
1 Fixed shade	F2	Cactus ribs change exposure.	Projecting ribs shade the façade.
2 Conductive fins	F3	Elephant hairs act as small fins.	Separated projections conduct heat to air.
3 Surface texture	F3	Sparse hairs suggest distributed structures.	Small roughness elements alter near-wall airflow; benefit is a hypothesis.
4 Solar reflector	F1	Silver-ant hairs reflect sunlight.	A reflective coating reduces solar absorption on a façade or roof.
5 Infrared emitter	F3	Silver-ant hairs enhance thermal emission.	A sky-facing surface releases infrared heat.
6 Fluid transport	F3	Blood carries heat to jackrabbit ears.	A fluid carries heat to a separate exchange surface.
7 Variable exchange area	F3	Ear blood flow regulates heat release.	Connect more exchange area when cooling demand rises.
8 Responsive shade	F2	Unequal swelling bends pine-cone scales.	A humidity-responsive layer changes shade position.

3.2.2 Combining Ideas with a Morphological Matrix

Table 3.2 groups alternatives by function. Bracketed numbers refer to Table 3.1; Option 3 draws on further models from Section 3.1. Choose compatible options for the functions a concept addresses.

Table 3.2 – Alternatives for heat shielding, reflection, heat transport and heat release.

Design choice	Option 1	Option 2	Option 3
S – Heat shielding	Fixed ribs (1)	Responsive louvers (8)	Insulating outer layer (camel fur)
O – Solar reflection	Reflective roof (4)	Reflective façade (4)	Multilayer reflector (Morpho wings)
T – Heat transport	Fluid loop (6)	Conduction through texture (2/3)	Fan-driven air cavity (bee fanning)
E – Heat release	Roof emitter (5)	Wall exchange with local air and sky	Evaporating wet layer (camel sweat)

A = S1–O1–T1–E1 combines fixed fins, a reflective roof and a water loop. Variable connected area (Crazy 8s, Idea 7) remains an unselected regulation option for **A**.

B = T2–E2 uses a textured coating for local heat exchange.

C = S2–O2–E2 combines responsive louvers with a reflective façade.

The strategies listed under Option 3 in Table 3.2 were not selected because of water or power demand, uncertain optical performance, or insulation restricting outward heat transfer. Fans and cladding are conventional technical implementations.

C was set aside because humidity does not directly indicate cooling demand and reflective façades also increased pedestrian heat stress in simulations of a Stuttgart street canyon [18]. **A** therefore places its reflective surface on the roof.

A and **B** were retained for the engineering comparison below.

3.2.3 Concept A V1 — Water Façade

A V1 was conceived for a new building, with water pipes embedded in the façade and timber fins providing shade (Figure 3.6). The water loop carries residual heat to a separate roof surface for release.



Figure 3.6 – Concept A V1. Timber ribs shade the façade; water carries residual heat to the roof and returns after cooling (schematic illustration).

Cactus ribs [10] inspire shading while the blood flow to jackrabbit ears inspires heat transport to a separate exchange surface [7]. The pipes and thermosiphon are conventional engineering choices: warmer water can rise and cooler water return if buoyancy overcomes pipe resistance [19].

3.2.4 Concept B V1 — Roughness Spray Retrofit

B V1 is a retrofit coating for an existing wall. Elephant hairs suggest small heat-transfer structures, but work as conductive fins [8]. The spray instead aims to disturb warm near-wall air (Figure 3.7). Its benefit depends on texture size, spacing, conductivity and local wind.

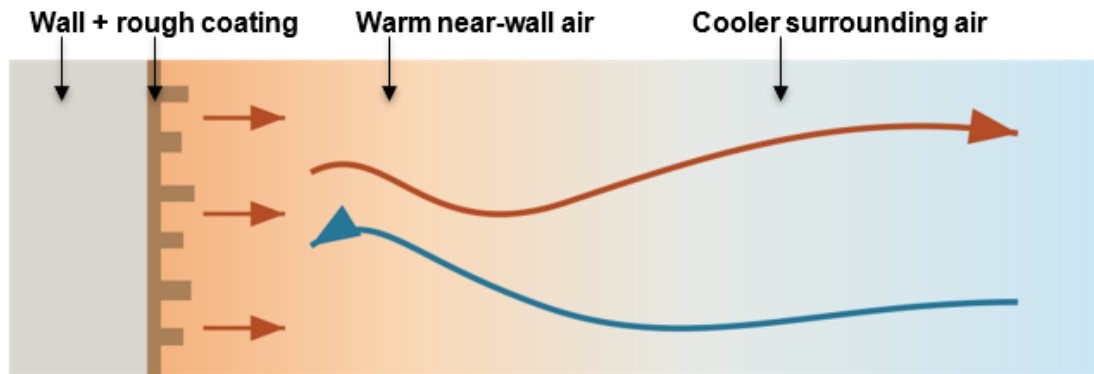


Figure 3.7 – Proposed roughness mechanism. Orange: warm near-wall air (thermal boundary layer); blue: cooler surrounding air, with a gradual temperature transition. Straight arrows: heat from the wall. Curved arrows: proposed air exchange.

3.2.5 Engineering Canvas

Table 3.3 summarizes the two V1 concepts in the Engineering Canvas.

Table 3.3 – Engineering Canvas for the two V1 concepts.

Canvas item	A V1 — water façade	B V1 — roughness spray
Function	Prevent heat gain; transport and release residual heat.	Increase local heat exchange with a thin retrofit.
Biological principle	Cactus shading; blood-borne heat transport; silver-ant optical properties.	Sparse elephant hairs provide distributed heat-transfer structures.
Technical transfer	Shade, conduct heat into water, circulate by buoyancy, release it at a reflective roof emitter.	Conduct heat into a texture and test its effect on airflow.
Components / materials	Timber ribs, collector, pipes, roof plate/coating, insulated return, expansion vessel and vents.	Existing wall, coating binder and textured particles; chemistry remains open.
Inputs / outputs	Sunlight and temperature difference → reflected light and roof heat release. Water recirculates.	Sunlight and airflow → heat exchange with local air and sky.
Requirements / KPIs	Section 2 KPIs; little auxiliary energy; winter operation, glare and townscape constraints.	Same KPIs; thin application, adhesion and safe weathering.
Critical assumptions	Enough heat reaches the water; roof capacity and buoyancy are sufficient.	Texture improves heat transfer without trapping still air.
Uncertainties	Pipe resistance, roof output, leakage, frost and panel-to-pipe contact.	Cooling benefit, soiling, adhesion and weathering.
Next check	Estimate circulation and roof capacity.	Compare smooth and textured samples under the same wind and temperatures.

3.2.6 Critical Engineering Questions

Q1: Can buoyancy circulate enough water? Q2: Can the roof release the collected heat? Q3: Does the spray improve heat transfer under weak airflow? These questions test the assumptions most likely to change concept selection.

3.2.7 Engineering Check 1 — Circulation and Pipe Size

The first estimate applies **A V1** to a hypothetical new building with the dimensions of the Modellierungsgebiet 03 reference block. Table 3.4 lists the model inputs and the engineering assumptions made by the project team.

Table 3.4 – Assumptions for the circulation estimate.

Input	Value and basis
Building and façade	64 × 58 m outer plan; 42 × 36 m courtyard; eaves at 15.4 m and roof rise 4 m. Two faces, assumed 65 % opaque: Area = (64 + 58) × 15.4 × 0.65 ≈ 1,220 m ² .
Collected heat	Assume $q'' = 100 \text{ W/m}^2$ enters the water after shading; test 50–150 W/m ² . Collection depends on panel-to-pipe contact.
Height H	Assume 10 m between centres of heat collection and release.
Water near 30 °C	Density $\rho \approx 996 \text{ kg/m}^3$; heat capacity $c_p \approx 4,180 \text{ J/(kg·K)}$; expansion coefficient $\beta \approx 3.0 \times 10^{-4} \text{ K}^{-1}$ [20, Table 3]. Assume viscosity $\mu = 0.8 \text{ mPa·s}$.
Common pipe route	Assume smooth pipe, $L = 160 \text{ m}$, fitting-loss factor $\Sigma K = 20$. Extra branch and radiator losses are excluded.

The chosen water-temperature differences, 10 °C and 3 °C, test stronger and weaker driving conditions; they are not measured site values or comfort targets. A smaller difference weakens buoyancy while requiring more flow to carry the same heat load.

Heat rate $\dot{Q}$ is energy carried per second; $\dot{m}$ is water mass flow. Assume steady liquid flow and approximate the hot/cold leg difference by the water temperature change ΔT . Equations use kelvin (K): a difference of 10 K equals a change of 10 °C.

$$\dot{Q} = A_f q'' \approx 1,220 \text{ m}^2 \times 100 \text{ W/m}^2 = 122 \text{ kW} \quad (3.1)$$

$$\dot{m} = \frac{\dot{Q}}{c_p \Delta T}; \quad \Delta p_b \approx \rho g H \beta \Delta T \quad (3.2)$$

$$\Delta p_{\text{loss}} = \left(f \frac{L}{D} + \Sigma K \right) \frac{\rho v^2}{2}; \quad v = \frac{4\dot{m}}{\rho \pi D^2} \quad (3.3)$$

Here D is internal diameter, v water speed and $g = 9.81 \text{ m/s}^2$. Use Darcy friction factor $f = 0.3164/\text{Re}^{0.25}$, with $\text{Re} = \rho v D / \mu$, and solve where friction equals buoyancy [19] (Table 3.5). $\text{Re} \approx 26,000\text{--}39,000$ supports the smooth-pipe turbulent-flow approximation.

Table 3.5 – Circulation estimates for the 122 kW load.

Water difference	Buoyancy pressure	Water flow	Internal diameter
10 °C (= 10 K)	≈300 Pa	2.9 kg/s	≈180 mm
3 °C (= 3 K)	≈90 Pa	9.7 kg/s	≈400 mm

Reducing the temperature difference more than doubles the required main-pipe diameter. These estimates use most of the available buoyancy pressure before radiator losses are included. **A** therefore needs less connected façade area, larger mains or shorter parallel circuits. Varying the assumed heat input from 50 to 150 W/m² gives a load of 61–183 kW.

3.2.8 Engineering Check 2 — Roof Heat Release

In steady operation, the roof must release the heat collected by the façade. Raman et al. [21] measured 40.1 W/m² net cooling at air temperature under sunlight above 850 W/m². This specialised emitter provides a benchmark, not an assigned property of our coating.

$$A_{\text{release}} = \frac{\dot{Q}}{q''_{\text{net}}} \quad (3.4)$$

For the engineering estimate, assume a 35 °C roof, 30 °C air, effective sky temperature 20 °C, solar reflectance $r = 0.95$, thermal emittance $\varepsilon = 0.90$, sunlight $G = 800 \text{ W/m}^2$ and convection coefficient $h = 3\text{--}8 \text{ W/(m}^2\cdot\text{K)}$. Effective sky temperature represents incoming infrared radiation, not air temperature. Neglect back-surface exchange and assume an unobstructed sky:

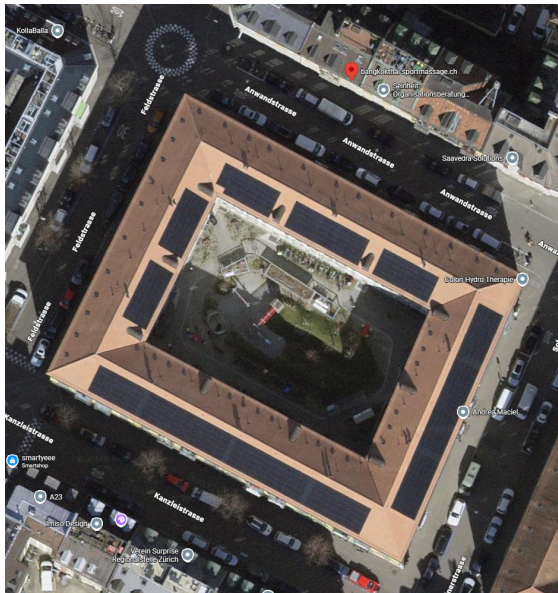
$$q''_{\text{net}} = h(T_{\text{roof}} - T_{\text{air}}) + \varepsilon\sigma(T_{\text{roof}}^4 - T_{\text{sky}}^4) - (1 - r)G \quad (3.5)$$

The terms represent heat to air, infrared heat exchange and absorbed sunlight. Fourth powers require absolute temperature: $T[\text{K}] = T[^\circ\text{C}] + 273.15$; thus 35 °C = 308.15 K. With $\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2\cdot\text{K}^4)$, net release is 15–40 + 83 – 40 $\approx$ 58–83 W/m² (Table 3.6). Roof tilt and nearby buildings would change this estimate.

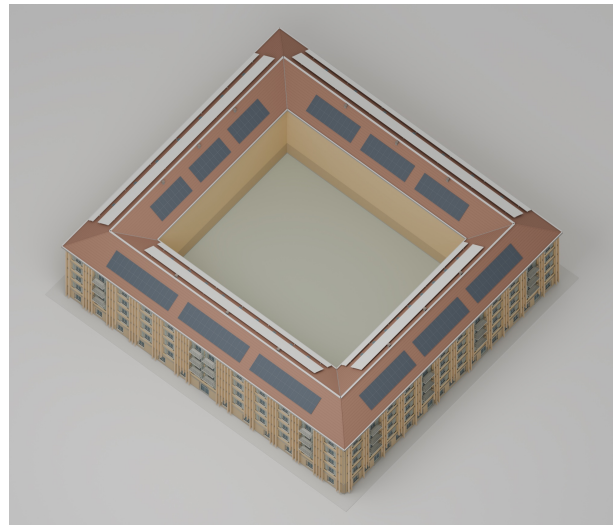
Table 3.6 – Roof cooling area required to release the assumed 122 kW load.

Case	Net heat release	Roof area for 122 kW
Published emitter benchmark	40.1 W/m ²	$\approx 3,050 \text{ m}^2$
Assumed 35 °C roof, 20 °C sky	58–83 W/m ²	$\approx 1,470\text{--}2,100 \text{ m}^2$
Same roof, warmer 30 °C sky	4–29 W/m ²	$\approx 4,200\text{--}29,500 \text{ m}^2$

The model allocates about 740 m² to cooling: 168 m total strip width $\times$ 4.39 m combined slope depth. Solar panels, rooflights and gaps limit this area (Figure 3.8). The first two cases give about 30–62 kW, supporting roughly 300–620 m² of façade at 100 W/m². A warmer sky reduces radiative heat loss; near-zero net output makes the required area highly sensitive. The connected façade area must therefore be reduced. Heat released to air remains local heat redistribution.



(a) Reference building — Google Maps



(b) Proposed roof layout

Figure 3.8 – Left: reference block in Modellierungsgebiet 03 [2], [3, pp. 148–151]. Right: our model with cooling strips sharing the roof with solar panels and rooflights.

3.3 Concept Evaluation & Iteration

3.3.1 Comparison against the KPIs

Table 3.7 compares the concepts using the estimates from Sections 3.2.7–3.2.8. Surface-temperature reduction has been calculated. Assumptions and untested quantities remain explicit.

Table 3.7 – KPI comparison using preliminary engineering estimates.

Criterion / unit	A V1 — water façade	B V1 — roughness spray
Solar reflectance / 0–1	Roof: 0.95 assumed	Not specified
Added shade / min	Fins added; duration not calculated	0 min added by the coating
Heat release / kW	740 m ² roof: ≈30–62 kW across benchmark / cooler-sky cases; ≈3–22 kW for warmer sky	Not quantified
Heat transport / kW; kg/s	122 kW assumed load requires 2.9 kg/s at 10 °C difference or 9.7 kg/s at 3 °C	Local exchange; rate not quantified
Intentional evaporation / L/h	0 in normal operation; filling and leakage are separate	0 for a dry coating
Performance retained / %	No ageing data	No ageing data

The flow rates are requirements, not demonstrated circulation. Even the highest roof estimate is below the assumed 122 kW load. Missing targets and test data prevent a defensible weighted score; the comparison instead identifies the changes needed for **A** and the evidence still needed for **B**.

3.3.2 Engineering Review and Selection

Table 3.8 lists the issues raised in the engineering review and the team’s response to each.

Table 3.8 – Team review and critical assessment of AI feedback.

Review issue	Response	Resulting decision
Embedded pipes risk hidden damage	Accepted	Move pipes into a separate external layer; enable access and retrofit.
Roof area and resistance limit A	Accepted	Limit connected area; use larger mains and short parallel circuits.
Weak ventilation rules out B	Qualified	Airflow is uncertain, but texture does not automatically fail. Test its mechanism.
Existing KPI definitions are incomplete	Accepted	Use heat-flow units; define targets and test conditions before measurement.

Reduced air exchange in Modellierungsgebiet 03 makes **B**’s airflow-dependent benefit uncertain [2], [3, pp. 148–151], [22]. Elephant hairs can aid heat transfer at low wind, whereas transverse wooden ribs reduced natural convection in another study [8], [23]. These contrasting results support testing the coating rather than assigning it a cooling value.

A was selected because it combines shading with heat transport to a separate release surface. Its greater material and installation needs remain a trade-off. The review in Table 3.8 guides the V2 construction changes and the revised engineering assessment below.

3.3.3 Final Concept V2 — Construction and Heat Paths

V2 moves the pipes outside the original wall, allowing both new construction and retrofit. The wall is retained apart from mounting connections. A small gap behind the added layer is intended to let escaped water fall away externally, reducing hidden wall-damage risk without guaranteeing leak-proof protection (Figure 3.9).

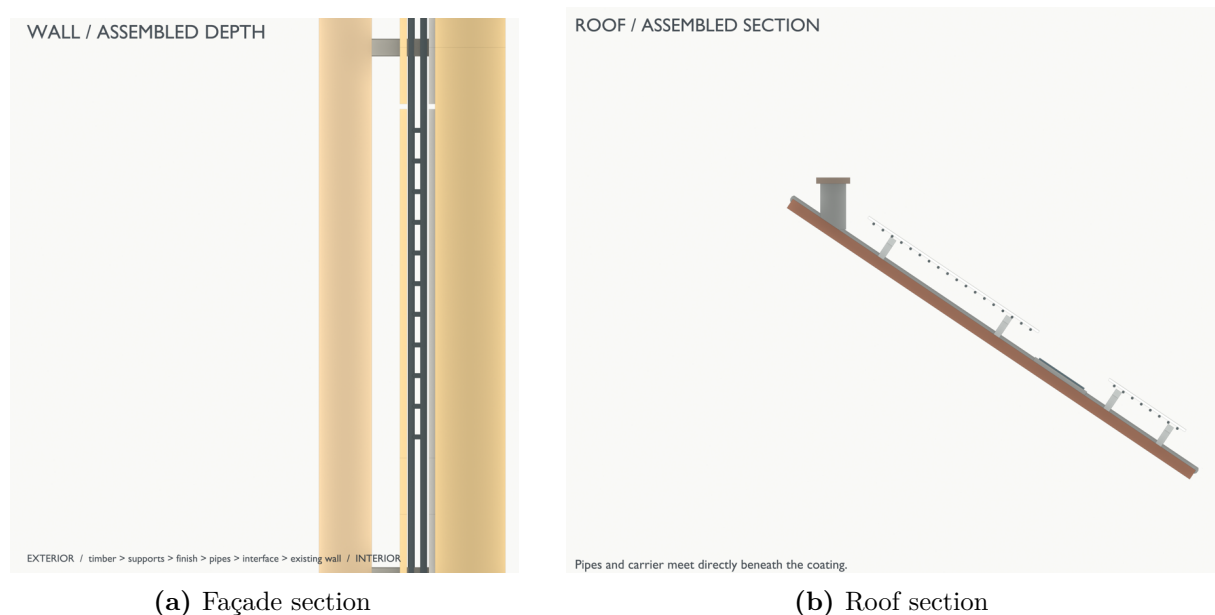


Figure 3.9 – Our assembled sections. Façade, outside inward: timber ribs, supports/air space, collector panel, pipes, interface, existing wall. Roof: coating, carrier plate, pipes, supports, existing roof.

Heat passes from the façade panel into water, then from the roof pipes into the carrier plate and coated surface (Figure 3.10). Good pipe-to-panel contact is essential. The Saharan silver ant inspires solar reflection and infrared emission [24]. The assumed 95 % reflectance still needs material verification. No increase in convection is assumed and convective heat release requires a surface warmer than the air.

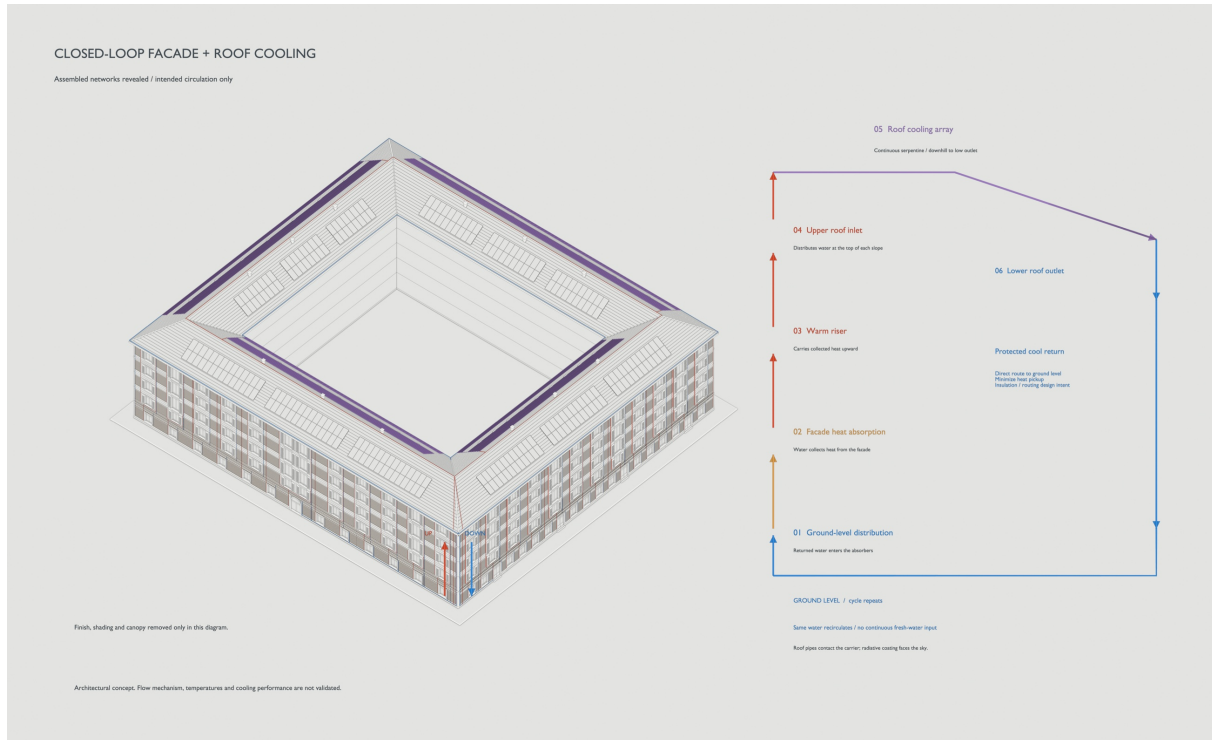


Figure 3.10 – Intended heat-transfer loop: façade collection → warm riser → roof heat release → cool return.

3.3.4 Pedestrian Space and Maintenance Access

Compared with V1, we cut back the timber fins to start at the first floor, about 3.4 m above ground (Figure 3.11). A new building could set its wall back to accommodate the fins; this retrofit must retain the wall beside the pavement. Reducing the modelled ground-floor projection from 0.66 m to 0.20 m leaves 0.46 m more space. Door operation and remaining pavement width still need checking.



(a) Fins start at the first floor



(b) Thinner base beside the pavement

Figure 3.11 – Revised timber fins and pedestrian space in our model.

Removing individual panels exposes the pipes for inspection and repair without opening the original wall (Figure 3.12). This improves access compared with V1's embedded pipes. Panel fixings and working clearance still need detailing.

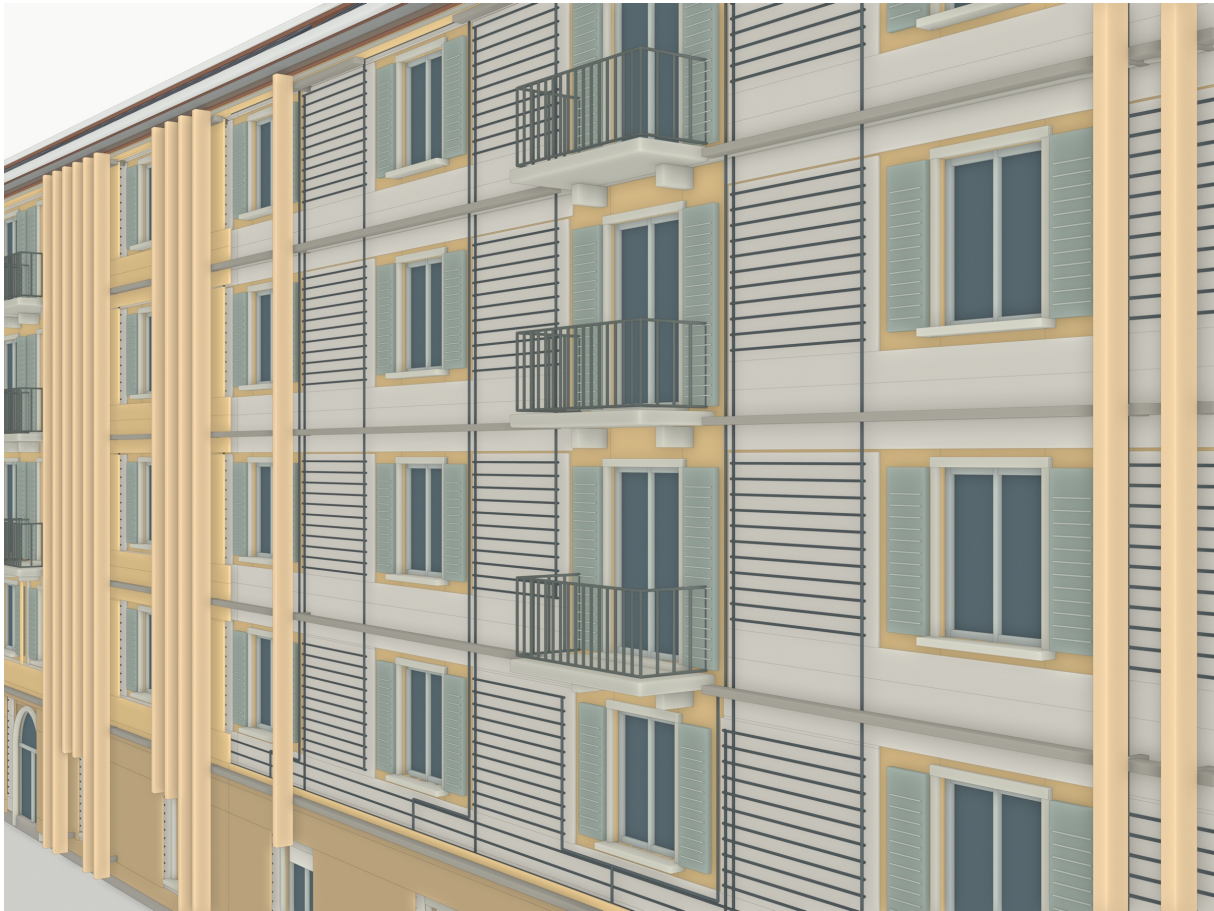


Figure 3.12 – Pipe-access detail from our model (view 17). Parts of the covering are removed to reveal the pipes behind the proposed separable façade panels.

Heritage protection remains a constraint [2], [3, pp. 148–151]. Retaining the wall and demonstrating a cooling benefit could support the proposal, but its appearance and attachments still require assessment.

Removable components support repair and material separation. Rib spacing should balance summer shade with winter sunlight. Timber treatment, coating durability, chemical safety and glare remain material checks. Reflective roofs may increase winter heating demand [25].

3.3.5 Updated Engineering Assessment

For an illustrative roof output of 60 W/m^2 , the 740 m^2 cooling area releases about 44 kW , supporting roughly 440 m^2 of façade at 100 W/m^2 . Repeating the circulation check with the same height and common route gives Table 3.9.

Table 3.9 – Pipe-size estimates after reducing the connected heat load.

Water difference	Buoyancy pressure	Water flow at 44 kW	Internal diameter
10°C (= 10 K)	$\approx 300 \text{ Pa}$	$\approx 1.1 \text{ kg/s}$	$\approx 120 \text{ mm}$
3°C (= 3 K)	$\approx 90 \text{ Pa}$	$\approx 3.5 \text{ kg/s}$	$\approx 255 \text{ mm}$

The drawn roof uses four long coils. Area divided by 0.16 m pipe spacing gives about $740/0.16 \approx 4,620 \text{ m}$ of pipe, or $1,150 \text{ m}$ per coil. Assuming 40 mm internal diameter, equal flow and a 10°C water difference, the Darcy equation gives about 20 kPa loss per coil at 44 kW total load, excluding bends. This exceeds the 0.30 kPa buoyancy pressure by roughly a factor of 67: the long-coil arrangement must change.

The roof and water-flow estimates prescribe temperatures separately. Water temperature, flow and pipe-to-surface heat transfer must be solved together before assigning an operating output. The rendered long coils therefore remain a layout to revise, with short parallel circuits and larger connecting pipes.

Roof-pipe water volume is approximately $4,620 \text{ m} \times \pi \times (0.04 \text{ m})^2/4 \approx 5.8 \text{ m}^3$, excluding façade pipes and headers. This makes roof loading and winter water storage important checks.

Winter operation. Before frost, the proposed circuit is emptied into storage and refilled when needed. Drainback requires sloping pipes and air entry; Baechler and Love [26, Ch. 3] give about 2.1% slope. Trapped water and pump-assisted refill remain checks.

3.3.6 Priorities for Further Development

Table 3.10 ranks the checks needed to develop Concept V2 further.

Table 3.10 – Ranked checks needed to develop Concept V2.

Priority	Check and design decision
1 Circulation	Size the parallel network including all losses; test start-up at small temperature differences. Decide whether buoyancy is sufficient.
2 Roof output	Measure heat release under varied sun, wind and sky conditions. Set the connected façade area from the result.
3 Winter operation	Test emptying and refill of one representative branch; measure retained water before freeze–thaw testing.
4 Panel assembly	Check pipe contact, leak behavior, removable connections and mounting loads.
5 Building and materials	Check pedestrian clearance, heritage compatibility, glare, winter gains, weathering, chemical safety and repair.

4 Sustainability & Life's Principles

4.1 Life's Principles applied to Concept V2

4.1.1 Be resource efficient — minimize the use of external resources

Cooling needs no auxiliary electricity: circulation is driven by buoyancy alone (Section 3.2.7), and the loop is closed, so no water is consumed in operation (Table 3.7). Pumpless operation is not demonstrated. The mains alone use most of the ≈ 300 Pa buoyancy pressure, the drawn roof coils exceed it by a factor of about 67, and refilling after drainback may need a pump (Section 3.3.5).

4.1.2 Adapt to changing conditions — day/night and seasonal cycles

The daily cycle is used rather than fought: a hotter façade produces a larger density difference and more flow, so cooling rises and falls with the load without a controller (Section 3.2.7), and the roof releases most of its heat to the night sky (Section 3.2.8). The seasonal cycle is met by draining the loop before frost and refilling it in spring (Section 3.3.5). The seasonal adaptation therefore functions as a withdrawal rather than an adaptation. Even after draining the pipes there is a remaining risk of freezing leftover water. Furthermore the ribs and the reflective roof act the same even in winter.

4.1.3 Use life-friendly chemistry — no toxic chemicals or materials

The heat-transfer fluid is plain water. Frost protection is solved by draining the pipes instead of antifreeze such as glycol. Reflection and emission are sought through the surface optics of the silver ant [24], not through a functional additive. We meet this principle by leaving substances out; we have not verified the ones we use. Two points remain open and unchecked: the timber treatment of the ribs and the binder and preservatives of the roof coating.

4.2 Environmental benefits, limitations and trade-offs

4.2.1 Environmental benefits

The climate benefit is what V2 replaces: every hour of air conditioning it makes unnecessary is avoided emissions and avoided waste heat in the street (Section 2.1). Radiation to the night sky carries the collected heat out of the urban air. The other release options of Table 3.2, exchange with the air beside the wall or evaporation, leave it in the street (Section 3.2.8). The street itself should be cooler. The shaded façade radiates less onto the pavement, there is no condenser at pedestrian height, and reflection happens on the roof, where it does not raise heat stress in the canyon the way a reflective wall does [18]. The retrofit is reversible, nothing is demolished. The panels can be removed and the building returned to its previous state, so the retrofit creates no demolition waste and consumes no land. In operation the system consumes no water and contains no refrigerant; what it does release is the abrasion of the roof coating into the run-off (Section 4.3).

4.2.2 Limitations

The export of heat depends on a clear, dry night sky. In the humid, hazy nights of a heat wave most of the heat only warms the roof air, so the concept helps least in the conditions it is meant for (Section 3.2.8). Outside the summer the drained loop does nothing, and no other environmental function fills the gap. No cooling figure exists, so the avoided emissions cannot be counted, and the whole benefit rests on circulation without a pump, the first open check of Table 3.10. The added materials carry embodied emissions that the constraint in Table 2.1 asks

to be repaid within the service life; we carried out no life cycle assessment, so neither side of that comparison exists.

4.2.3 Trade-offs

Ribs and panels change how the building looks. On a listed façade this collides with heritage protection [2], [3, pp. 148–151]. What shades and reflects in summer also rejects winter sun and can raise heating demand and its emissions [25]. The net annual effect is unknown. Every square meter of roof that releases heat is a square meter without photovoltaic yield (Figure 3.8). Avoiding antifreeze costs the shoulder seasons and admits air that may bring back a corrosion inhibitor (Section 3.3.5). Repairable external pipes are the same visible layer that alters the façade.

4.3 Materials, durability, repair and circularity

The timber ribs are worn by UV, driving rain and freeze–thaw, so they belong to the biological cycle [27]. What weathers off lands on the pavement, and the timber must be untreated, naturally durable, or protected by detailing rather than by chemistry. The collector panels and their finish see slow surface wear and are otherwise used as structure: technical cycle, if the panel is one recoverable material with a benign finish. The panels are designed to be removable (Figure 3.12). The roof coating is worn by UV, rain, soiling and cleaning, and abraded into run-off that reaches the stormwater network of Section 2.3, so it must break down harmlessly. Pipework, carrier plate, supports and fixings are used, not consumed: technical cycle, recoverable if specified as separable single materials. The water in the loop is used but mobile and fits neither cycle cleanly. About 5.8 m³ is drained to storage each winter and eventually disposed of, and with an inhibitor it is the one component that can leave the system by accident. Measured against the eight cycle design principles of the lecture [28], V2 answers four in principle and none with evidence. Modular construction (3) and reparability (5) follow from moving the pipes out of the wall: a panel can be removed and the façade pipes inspected without opening the wall (Sections 3.3.2 to 3.3.4). Ecotoxic chemicals (1) and recyclable materials (2) are answered above.

Material checked so far. White cool-roof coatings owe their reflectance to titanium dioxide, so it was checked in the REACH database of the European Chemicals Agency [29]. The powder form was classified in 2020 as a suspected carcinogen by inhalation (Regulation (EU) 2020/217, H351). The General Court annulled the classification in 2022 and the Court of Justice confirmed the annulment on 1 August 2025 [30], so the pigment currently carries no harmonized hazard classification. The result is not clear. The concern was dust inhalation, not the applied film, and the ruling turned on the reliability of one study, not on a finding of safety. The pigment is also not the used coating. The binder and the film preservatives are what reach the run-off instead. One substance of at least four has been checked. The other three belong to item 5 of Table 3.10.

4.4 Did sustainability considerations change the design?

The design was adjusted in four instances which were documented in Sections 3.2 and 3.3.

1. The reflective surface moved from the façade to the roof. Concept C reflected on the façade and was set aside because reflective walls raised pedestrian heat stress in simulations of a Stuttgart street canyon [18]; Concept A reflects on the roof instead (Section 3.2.2). The change was made for the people at street level, not for cooling performance.

2. Water and power demand removed two options before any concept was drawn. The evaporating wet layer and the fan-driven cavity of Table 3.2 were not developed further because of their water and power demand (Section 3.2.2). The Energy and Water constraints of Table 2.1, which are the principle “be resource efficient” written as constraints, decided this.

3. The pipes moved out of the wall. V1 embedded them in the façade. The engineering review found that in case of damage the lack of access was unacceptable (Table 3.8). V2 therefore placed them in separable external panels with a drainage gap behind them (Section 3.3.3).

4. Water instead of glycol. Frost protection was decided by drainback rather than by an additive (Section 3.3.5).

4.5 Limits of this assessment

The assessment is made against Concept V2 as described in Section 3.3.3, whose central claim of buoyancy-only circulation is the first-ranked open check of Table 3.10. If a pump is needed, the Energy constraint and the resource-efficiency assessment of Section 4.1 fall with it. Cascade use and return logistics lie outside the project boundary, and of the substances the concept would put into the street one has been checked and three have not.

5 Discussion & Conclusion

5.1 Contribution of the biomimicry process

Biomimicry helped us explore cooling functions before committing to familiar building solutions. Studying how organisms manage heat encouraged us to abstract and combine strategies, leading to the proposed link between façade shading, fluid transport and a roof emitter (Sections 3.1–3.2). Without this process, we would probably have focused on improving existing solutions individually. Conventional engineering could produce a similar concept, but the biological models broadened our search and helped us connect these functions.

5.2 Strengths and weaknesses of the final concept

V2 offers an unconventional approach for new buildings and especially retrofits: shading is intended to limit façade heating, while the water loop transfers collected heat to the roof. This aims to reduce heat release at street level without relying on reflective façades that could increase exposure for pedestrians and neighboring buildings. Modular panels could be installed where cooling is most needed. Heat-driven circulation and a closed loop offer the prospect of operation without continuous pumping or intentional evaporative water use. However, circulation and the available roof area are the main limitations. The calculated pipe resistance exceeds the available driving pressure, requiring a revised network with shorter parallel circuits (Section 3.3.5). Pump-free circulation remains unverified, and refilling may require a pump. Testing must establish whether the revised system delivers the intended cooling.

5.3 Assessment of the challenge and the KPIs

The concept addresses the façade’s contribution to urban heat; it does not address all causes of the wider challenge. The KPIs guided design decisions: heat-flow estimates limited the connected façade area, and the water constraint favored a loop without intentional evaporation (Section 3.3). They have therefore supported concept development, but target achievement has not been demonstrated. Reflectance, shading duration, durability and actual cooling performance still need verification. Winter operation, glare and heritage compatibility also remain open. These gaps define the next evaluation steps.

5.4 Innovative character and limits of the evidence

The innovative aspect is the combination of biological principles in a maintainable façade system intended for new buildings and retrofits. We do not claim that its components are new. Our main lesson from the course is that biological ideas become more useful when calculations and practical constraints shape their technical transfer. The estimates support further development, but do not establish measured cooling performance. Circulation and roof-output tests are the priorities for assessing the concept’s potential.

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Appendix A – AI Use Statement

Relevant AI-supported activities are documented in Table A.1. Not every prompt is listed; the focus is on how AI contributed to the work and how important outputs were verified. AI output is not used as evidence for factual claims.

Table A.1 – AI use statement.

AI tool	Used for	How the output influenced the project	How important output was verified
AskNature (search, strategy pages and AskNature Chat)	Discovering biological strategies for functions F1–F5 with the keywords and biologized questions of Section 3.1.1	Produced the candidate list of natural models in Section 3.1.2, including the models that were subsequently de-selected	Every mechanism claim carried into the report is referenced. Each strategy page was opened and read; the access dates are given in the reference list. Where a primary paper exists it is cited alongside or instead of the AskNature page (elephant hair, cactus ribs, silver ant, Morpho wing, pine cone, honeybee)
Large language model assistant (Claude, Anthropic)	Structuring the scoping notes; drafting and editing report text; converting the Word draft to L ^A T _E X; an editorial audit of the draft against the course rubric (repetition, unsupported claims, citation and numbering gaps)	Draft wording in Sections 1–5 and the layout of most tables; the audit produced the list of formal corrections applied in this version (figure and table numbering, missing references, citation style, headings)	The team decided on every audit finding. The numbers in Sections 3.2.7, 3.2.8 and 3.3.5 were recomputed by hand from the stated inputs before the text was accepted. Any claim the assistant proposed without a source was either given a source we read ourselves or removed
AI as critical reviewer (Claude, Anthropic, prompted as a critical engineering reviewer)	Reviewing Concept A V1 and B V1 against the Engineering Canvas, the functions and the KPIs; identifying the assumptions most likely to change concept selection (Section 3.3.2)	Four review issues in Table 3.8; three were accepted and changed the design (pipes into external panels, reduced connected area, KPI units), one was qualified	Each issue was assessed by the team against the literature named in Section 3.3.2; the qualified issue (ventilation) was not accepted as stated because the cited studies disagree
Image generation (imagine.art)	Schematic illustrations of the elephant ears and hair (Figures 3.1 and 3.2) and the Concept A V1 sketch (Figure 3.6) from prompts written by the team	Illustration only; no generated image is used as a measurement or as evidence. The renders of Concept V2 (Figures 3.8–3.12) are from the team's own 3D model	Each generated image was compared with the mechanism described in the cited source and labelled in the caption as schematic

On the limits of this statement. Nothing in this report is cited to an AI tool. Where an AI-assisted search surfaced a candidate, the candidate was carried forward only with a source of its own: the AskNature strategy pages are cited with their access dates, and where a primary paper exists it is cited in Section 3.1.2.

Appendix B – Optional Supporting Material

This appendix collects the glossary and the full-size drawings of the Concept V2 heat-transfer loop.

B.1 Glossary

Albedo Refers to an index from 0 – 1 on how much light is reflected.

PV Photovoltaic

Radiation Shortwave radiation is the type of radiation emitted by the sun. Longwave radiation is thermal radiation which is emitted from a surface into the surrounding atmosphere.

UHI Urban Heat Island

UV-Light Ultraviolet Light

B.2 Concept V2 heat-transfer loop

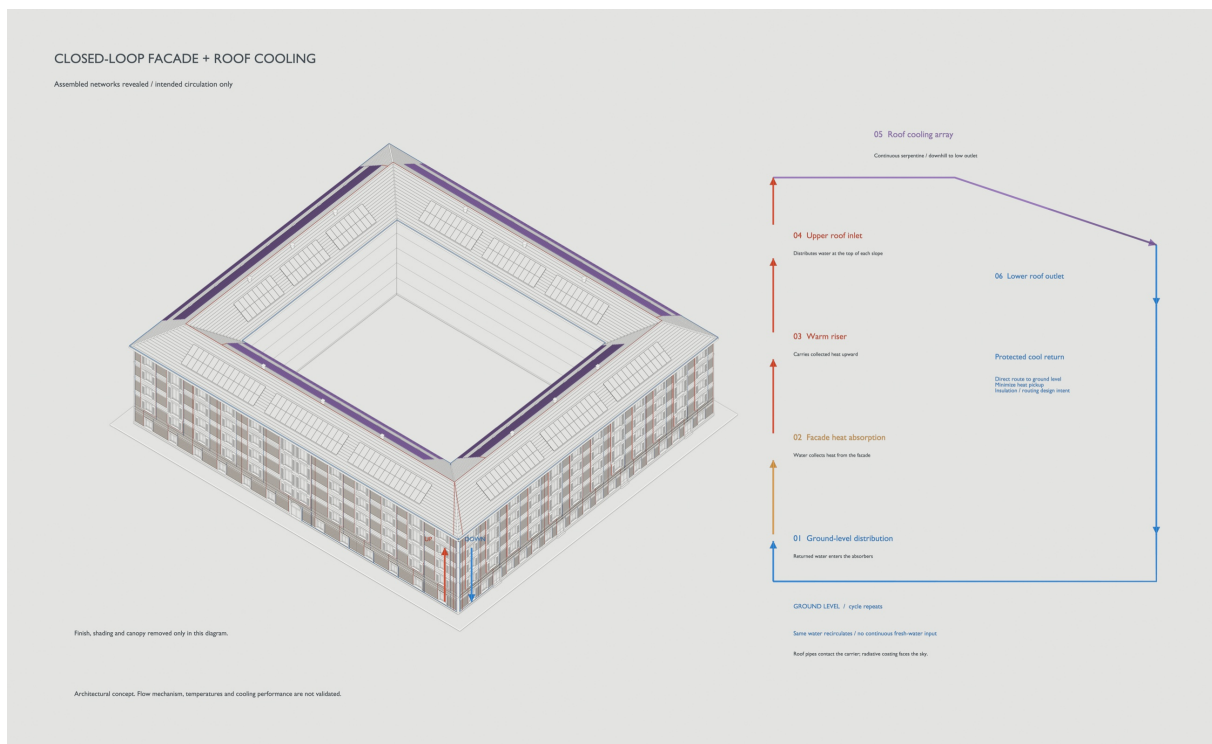


Figure B.1 – Closed-loop façade and roof cooling: assembled networks and intended circulation (full-size version of Figure 3.10).

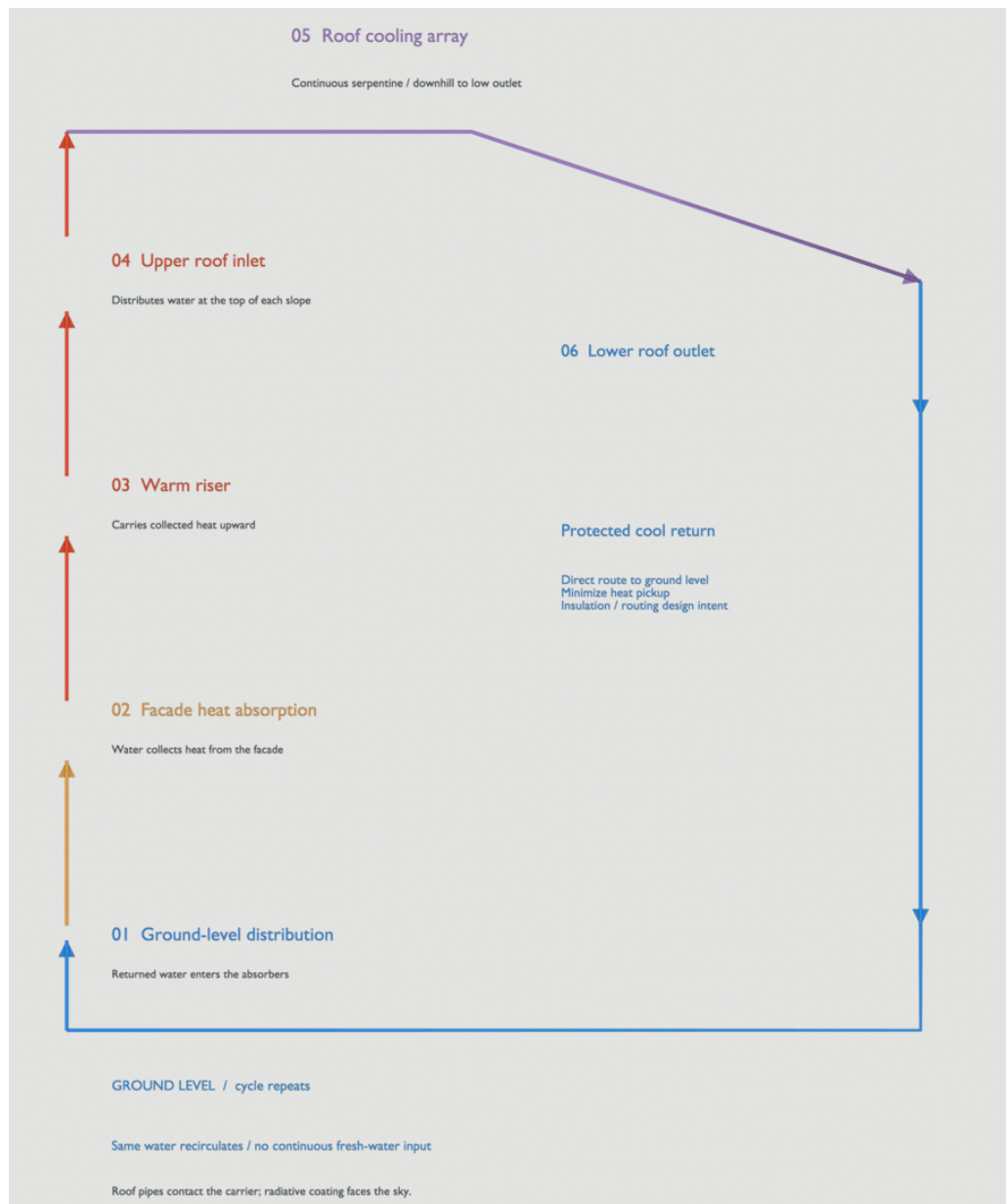


Figure B.2 – Flow schematic of the loop: ground-level distribution, façade heat absorption, warm riser, upper roof inlet, roof cooling array, lower roof outlet and protected cool return.